The Non-Canonical Frontier in Peptide-MHC-I Binding Prediction: A Benchmark-Hygiene Case Study

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Abstract

Motivation: Peptide-MHC-I binding-prediction benchmarks can manufacture false conclusions when confounders are uncontrolled. **Results:** We benchmarked six architectures on 5,190 MHCflurry-labelled HLA-A*02:01 9-mers with multi-seed estimates and re-analysed three apparent findings under controlled conditions. A convolutional network was modestly best (91.9% $\pm$ 0.9% vs. 90.6% $\pm$ 1.2% for a deep feed-forward network); random labels fell to chance ($\sim$33%); and a claimed "label quality dominates architecture" result dissolved once the degenerate null control was removed. A zero-parameter anchor rule matched the trained model on canonical peptides, while non-canonical binders plateaued at $\sim$0.65 sensitivity across three independent predictors — the true frontier. A temporal holdout of post-2022 epitopes preserved recognition, and a benchmark of 6,524 experimentally measured affinities (ROC AUC 0.942) confirmed the training labels' quality. We distil the artifacts into a benchmark-hygiene checklist applicable beyond peptide-MHC. **Availability:** Code, models, and data at github.com/woodhaha/peptide-mhc-benchmark. **Contact:** zhouzhuha@wmu.edu.cn; wzxxy@wmu.edu.cn **Supplementary information:** Supplementary data are available at *Bioinformatics* online.

1 Introduction

MHC class I molecules present 8–11-mer peptides to CD8+ T cells, and accurate prediction of peptide–MHC-I binding underpins epitope discovery for vaccine design and cancer immunotherapy (Falk *et al.*, 1991; Lundegaard *et al.*, 2011). For HLA-A*02:01 — the most prevalent class I allele globally — binding is dominated by two primary anchor positions: position 2 (p2, hydrophobic: Leu/Met/Ile/Val) and position 9 (p9, C-terminal, Val/Leu preferred) (Ruppert *et al.*, 1993; Rammensee *et al.*, 1995). Computational predictors have evolved from position-specific scoring matrices to neural networks and pan-specific tools such as NetMHCpan (Reynisson *et al.*, 2020), MHCflurry (O'Donnell *et al.*, 2020), BigMHC (Albert *et al.*, 2023), and MUNIS (Wohlwend *et al.*, 2025), increasingly using protein language model embeddings (Rives *et al.*, 2021).

Several gaps remain. Head-to-head architecture comparisons under fully controlled conditions (identical training data, encoding, evaluation) are uncommon. The impact of training-label quality on downstream performance is seldom quantified, despite evidence that $>$55% of IEDB entries are computationally rather than experimentally labelled (Preibisch *et al.*, 2026). And the translation of binding prediction to real-world epitope discovery is rarely demonstrated in a unified, reproducible pipeline.

We address these gaps through a systematic benchmark designed to isolate the effects of architecture choice, label quality, and encoding. Our contributions are: (i) a controlled six-architecture comparison with multi-seed uncertainty estimates, showing a modest but measurable architecture effect; (ii) a label-source control confirming that random labels fall to chance; (iii) evaluation of ESM-2 embeddings against BLOSUM62 with a mean-pooled ablation; (iv) held-out validation against 49 IEDB epitopes (44 of which overlap the labeler's training corpus); (v) translational application through protein scanning and cancer mutation analysis; (vi) a template-based structural hypothesis for non-canonical anchoring; and (vii) a multi-allele extension testing whether the recipe generalizes across common HLA-I alleles and how hard non-canonical binders are for three independent sequence-based predictors.

2 Methods

2.1 Training data and labels

100,000 random 9-mer peptides were generated by uniform sampling of the 20 standard amino acids. Each was labelled with MHCflurry 2.2.1 against HLA-A*02:01 and assigned to Strong Binder (SB, percentile rank $<$0.5%), Weak Binder (WB, 0.5%$\leq$rank$<$2.0%), or Non-Binder (NB, rank $\geq$2.0%) following the netMHCpan convention (Reynisson *et al.*, 2020). The resulting distribution (1.7% SB, 5.1% WB, 93.2% NB) was balanced by downsampling to 1,730 per class (n=5,190),

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partitioned 90/10 into training (n=4,671) and held-out test (n=519) sets with stratified sampling (seed 42). Two comparator labelling strategies were implemented: a PSSM encoding crystallographic A*02:01 anchor preferences, and random synthetic labels as a baseline control. A held-out benchmark of 49 experimentally validated IEDB epitopes (HLA-A*02:01-restricted, with positive binding/T-cell assay evidence) plus 20 homopolymer negatives ($\leq$2 unique amino acids) was curated from the IEDB (query-api.iedb.org, accessed 2026-08-02); label-circularity overlap (44/49) was determined by sequence-identity match against the IEDB entries in MHCflurry's training corpus (Vita *et al.*, 2019).

2.2 Peptide encoding

Each 9-mer was encoded with the BLOSUM62 matrix (Henikoff and Henikoff, 1992), min–max normalized, producing a 9×20 matrix flattened to 180 dimensions for fully connected networks. ESM-2 embeddings (Lin *et al.*, 2023) were extracted at t6 (2,880-dim per-peptide; 320 per position × 9) and t12 (4,320-dim; 480 per position × 9) depths — the two smallest ESM-2 checkpoints (8M and 35M parameters); a mean-pooled t6 variant (320-dim) served as an ablation control destroying positional information.

2.3 Model architectures

Six classifiers were implemented: (i) FFN (Nielsen *et al.*, 2003): 180→90→3, 49K params; (ii) Deep FFN: 180→360→180→90→45→3 with batch normalisation and dropout, 152K params; (iii) CNN: Conv2D(32,3×3) + FFN body, 1.05M params; (iv) LSTM(64)→dense(32)→3, 24K params; (v) ResNet-style stem + 3 residual blocks, 325K params; (vi) Random Forest (100 trees). For ESM-2 embeddings an FFN classifier with a 256→128→64 stack was trained. All networks used cross-entropy loss with Adam or RMSprop (lr 0.001), 150 epochs, batch size 50, 20% internal validation split, early stopping (patience 10), and learning-rate reduction on plateau. Five-fold stratified cross-validation was performed for the FFN. The held-out test set (519 peptides) was disjoint from training and used for all reported metrics.

2.4 Protein scanning, mutation analysis, and structure

Ten therapeutically relevant proteins (MART-1, gp100, tyrosinase, NY-ESO-1, WT1, p53, KRAS, influenza M1, CMV pp65, SARS-CoV-2 spike RBD) were scanned with a sliding 9-mer window (stride 1; 3,536 windows). Sixteen recurrent hotspot mutations (7 p53, 9 KRAS) were analyzed by comparing wild-type and mutant binding scores; mutations were classified as Created/Enhanced/Destroyed/Unchanged. The KRAS G12V neoepitope YKLVVVGAV carries a non-canonical lysine at P2; we evaluated a predicted K-E63 salt bridge using comparative B-pocket geometry across eight HLA-A*02:01 crystal structures (PDB 1DUZ, 1AKJ, 1HHJ, 1JF1, 1QEW, 1QRN, 1S9W, 2GIT), energy minimisation (OpenMM 8.5.2, amber14 force field), and a Coulombic electrostatic estimate. The structural argument here is deliberately restricted to minimisation and qualitative electrostatics.

2.5 Feature extraction

Learned features were extracted from the Deep FFN penultimate dense layer via truncated models. Hierarchical clustering (Ward's method, Euclidean distance) was applied to representations of top epitope candidates; feature–binding-score correlation was assessed by Pearson's r with a 1,000-iteration permutation test to control for chance correlations. t-SNE visualisation of the 45-dimensional penultimate layer was used to inspect class separation, and integrated gradients identified position-specific anchor contributions.

2.6 Multi-allele extension

Experimentally confirmed class I binders were curated from IEDB (Vita *et al.*, 2019) for five common alleles (HLA-A*02:01, A*01:01, A*03:01, B*07:02, B*44:03), yielding n=14,488 positives (canonical 11,225; Type A 151; Type B 3,112). Type A (tolerated weak anchors) and Type B (energy-compensated non-canonical) were defined by whether the peptide violates the allele's canonical anchor motif at p2/p9. Protein-background negatives were sampled from natural proteomes. A rebalanced pan-specific DFFN (B*07:02 upsampled 4×) was trained on pooled data with five seeds (0, 1, 42, 123, 2024). MHCflurry 2.2.1 (affinity $<$500 nM) and MUNIS (Wohlwend *et al.*, 2025) (ESM-2-based sequence model) served as independent comparators on the n=3,112 Type B population plus canonical controls. A 15-allele panel (pan16: 11 additional alleles) tested scaling with per-allele minimum sample (2,000) upsampling. To address the label-circularity concern, temporal holdout compared IEDB epitopes published after 2022 (post-2022, unseen by MHCflurry's ∼pre-2021 training) against pre-2021 (in-distribution) epitopes at sensitivity threshold 0.95. A broad-motif retraining control (extending the canonical anchor set to include tolerated residues) tested whether Type A under-recovery reflects missing training coverage.

3 Results

3.1 Training data and architecture comparison

MHCflurry labelling of 100,000 random 9-mers yielded 1.7% SB, 5.1% WB, 93.2% NB; after balancing, 5,190 peptides (1,730/class). Across five training seeds, the CNN achieved the highest accuracy (91.9% ± 0.9%; Table 1, Figure 1), modestly ahead of the Deep FFN (90.6% ± 1.2%), the FFN (89.8% ± 1.1%), and the LSTM (89.7% ± 0.9%) — the top four architectures within ∼2 percentage points — while ResNet (87.6% ± 1.8%) and Random Forest (79.7% ± 0.4%) were clearly worse. MHCflurry itself scored 100% on the test set it labelled — an expected ceiling, since the labels are its own predictions — so the meaningful comparison is the 8.1-point gap between the labeler's decision boundary and the best finite-sample model that approximates it (91.9% for the CNN); independent-label circularity is instead quantified by the 62.24% NetMHCpan agreement (see Discussion). The WB class was consistently the most difficult (F1 0.752–0.880).

The modest spread among the top four architectures reflects the positional nature of peptide–MHC binding: anchor positions p2/p9 dominate, so architectures that preserve positional information perform comparably. ResNet and the Random Forest were clearly worse, consistent with the absence of translation invariance in a 9×20 encoding and, for the Random Forest, the high dimensionality of the BLOSUM62 encoding.

3.2 Label-source controls

Label source, not architecture, sets the practical ceiling of what a model can learn (Table 2, Figure 2). Training on MHCflurry labels reached 90.6% accuracy, whereas a deterministic PSSM label source reached 82.6% (on a separate 15,000-peptide pool, so not a clean head-to-head comparison) and an i.i.d. random-label null fell to chance (34.5 ± 1.7%, close to the 33.3% expected for three balanced classes). The random-label result confirms the models learn genuine signal rather than noise; the PSSM–MHCflurry gap, though confounded by pool size, suggests MHCflurry's labels are at least as learnable as a hand-crafted scoring matrix.

3.3 Encoding

On the FFN (the architecture used for all encoding comparisons), BLOSUM62 generalized stably under cross-validation (92.2 ± 0.4%;

per-fold 92.2/92.4/92.1/92.7/91.6%). ESM-2 t6 per-position embeddings scored 93.3% on a single split but collapsed to 82.2 ± 1.0% under CV — the 2,880-dimensional features overfit the 4,671-sample training set (Table 3, Figure 7). Mean-pooled t6 embeddings collapsed to 65.9%, confirming that positional information is essential. The t12 variant (4,320-dim) overfit similarly (90.9–91.5%). The single-split 93.3% result reflects a favourable partition rather than a genuine encoding advantage at this training scale; at the data scale typical of single-allele studies, compact biologically motivated encodings such as BLOSUM62 remain the more robust choice, and ESM-2's representational advantage may require substantially larger training corpora to materialise.

3.4 Held-out validation and translation

The Deep FFN achieved 95.8% sensitivity (46/48), 75.0% specificity (15/20), ROC AUC 0.953 against IEDB epitopes (Table 5, Figure 3), after excluding ILRGSVAHK (influenza NP 265–273), which is HLA-B*08:01- rather than A*02:01-restricted. The two remaining false negatives (QYDPVAALF, TLGIVCPIC) carried non-canonical p9 anchors (F, C). An independent predictor, NetMHCpan-4.1 (queried through the IEDB web API, which does not pin a reproducible tool version), achieved 83.3% sensitivity (40/48) on the same set. This gap partly reflects label circularity (44/49 of the epitopes overlap MHCflurry's training corpus), but NetMHCpan's pan-specific training on MS-elution-based labels makes this an indicative rather than clean comparison; the more direct circularity evidence is the 62.24% label agreement (323/519 test peptides) reported in the Discussion. To contextualise the benchmark, we include a zero-parameter anchor-checker baseline (a rule classifying any 9-mer with canonical p2/p9 residues as a binder): it achieves sensitivity 1.0 on this canonical-motif-dominated set, matching the trained model (a comparison with only 3 discordant peptides, too low-powered for a formal test; the anchor rule has no tunable score threshold, so its precision—and hence F1—is fixed by the test-set composition rather than a free parameter) — confirming that the canonical set overstates generalisation and that the informative signal lies in the non-canonical/hard-tail sensitivity. The five false positives were homopolymers ($\leq$2 unique amino acids) with canonical anchors — biologically implausible binders whose low sequence complexity the classifier fails to penalise. A domain-knowledge filter ($\leq$2 unique amino acids $\rightarrow$ non-binder) removed all five, but we report 75.0% as the primary, unfiltered specificity to avoid post-hoc inflation. To break this circularity, we additionally scored the Deep FFN against a large label-independent set of experimentally measured affinities — 3,438 unique 9-mers with measured IC50 $<$500 nM and 3,086 with IC50 $>$5,000 nM (the 500–5,000 nM grey zone was excluded to yield unambiguous binders and non-binders), restricted to direct binding assays (competitive fluorescence/radioactivity) and excluding MS-elution-derived, predictor-dependent entries. The MHCflurry-trained Deep FFN recognised 94.4% of the experimental binders and rejected 74.5% of the non-binders (threshold 0.5; ROC AUC 0.942); the rebalanced per-allele model reached 89.8% sensitivity and 89.1% specificity (AUC 0.945). This benchmark validates the quality of the training labels — that MHCflurry's computational labels agree with experimentally measured binding — rather than generalisation to unseen peptides, since the peptides are drawn from the same IEDB binding-affinity corpus on which the labeler was trained; generalisation to unseen peptides is instead established by the temporal holdout (Section 3.6). Protein scanning of 10 proteins (3,536 windows) recovered 10/11 known epitopes (91%; Figure 4), including immunodominant GILGFVFTL (M1), NLVPMVATV (pp65), and SLLMWITQC (NY-ESO-1); the single miss was the MART-1 clinical variant ELAGIGILTV, whose P2 leucine is a non-natural anchor-modified residue. Mutation scanning of 16 hotspot mutations identified 7 epitope-altering events (Table 4, Figure 5). p53 R248W created the neoepitope MNWRPILTI, a mutation whose T-cell immunogenicity is documented in patients and in HLA-A2/DRB1-transgenic mice (Malekzadeh *et al.*, 2020; Quandt *et al.*, 2018); KRAS G12V created YKLVVVGAV via G$\rightarrow$V at p9 creating a canonical C-terminal anchor. Three mutations (KRAS G13D, KRAS A146T, p53 R249S) destroyed existing epitopes, representing potential immune-evasion mechanisms.

3.5 Structural hypothesis

Template-based analysis of eight HLA-A*02:01 crystal structures (Figure 8) showed a conserved B-pocket with Glu63 positioned 4.7 ± 0.2 Å from the P2 Cα. Energy minimisation refined the P2 Cα–Glu63 OE2 distance to 3.7 Å, within the reach of a lysine side chain (maximum extension 7.6 Å); note that a proper salt-bridge geometry requires the side-chain Nζ–OE2 distance rather than the backbone Cα, which we did not compute here because a complete lysine side chain was not modelled, so the Cα-based distances are a coarse geometric proxy only. An un-solvated Coulombic estimate would overstate the stabilisation (a buried salt bridge typically contributes only a few kcal/mol after desolvation). The K-E63 salt-bridge hypothesis therefore remains a qualitative, untested structural conjecture, pending experimental validation (e.g., X-ray crystallography or cryo-EM of the YKLVVVGAV–HLA-A*02:01 complex). In contrast, the p53 R248W neoepitope MNWRPILTI carries asparagine at P2, and its improved predicted binding relative to wild-type (MNRRPILTI) is attributable to the P3 Arg$\rightarrow$Trp substitution enhancing hydrophobic packing rather than any P2-mediated mechanism.

3.6 Multi-allele generalization

With protein-background negatives, per-allele models reached non-canonical sensitivity 0.84–0.96, and a rebalanced pan model matched them on four of five alleles (A*02:01 0.84, A*01:01 0.88, A*03:01 1.00, B*44:03 1.00; each per-allele estimate rests on n=25 non-canonical binders, so individual-allele parity is underpowered); B*07:02 was a notable exception (pan 0.60 vs per-allele 0.96; 0.64 in the 15-allele pan16 panel, Table 6), consistent with over-anchoring to P2-Proline (Figure 9). This corrects an earlier observation that per-allele models outperform pan models on non-canonical binders: that apparent advantage was a 10.6$\times$ training-data-size artifact, not biology. On the 15-allele panel, pooled non-canonical sensitivity was 0.819 (307/375), with 12/15 alleles at 0.76–1.0. The weak tail (B*27:05, B*51:01, B*07:02, A*24:02) reflected data scarcity and, for B*07:02, over-anchoring, not architecture. Temporal holdout (three alleles with sufficient post-2022 data) confirmed preserved recognition of epitopes published after the labeler's training cutoff at sens@0.95: A*02:01 0.824, A*03:01 0.936, B*07:02 0.973 (versus pre-2021 0.879/0.905/0.898), indicating the model is not overfit to a single training-data era.

3.7 Non-canonical plateau

Canonical binders were recognized at $\sim$0.98 by every predictor class: DFFN 0.979, MHCflurry 0.986, MUNIS 0.950. Non-canonical (Type B) binders plateaued at $\sim$0.65: DFFN 0.607, MHCflurry 0.668, MUNIS 0.647 (Table 7) — a second, ESM-2-based sequence model did not lift the ceiling. Type A (tolerated weak anchors) binders were inherently hard (DFFN 0.477, MHCflurry 0.669) and not recovered by broad-motif retraining (A*03:01 flat at 0.634; B*07:02 Type B dropped 0.680$\rightarrow$0.341). The extreme energy-compensated tail (e.g., KRAS G12V P2-Lys; P2-Lys/Arg/His sidechains, maximum extension 7.6/7.3/4.4 Å, span the 3.83 Å P2–Glu63 gap) was scored as non-binder by all

Table 1. Model performance on the MHCflurry-labelled test set (mean ± SD across five training seeds, n=519).

Model	Accuracy (%)	Parameters
CNN	**91.9 ± 0.9**	1,053,863
Deep FFN	90.6 ± 1.2	152,823
FFN (Jessen)	89.8 ± 1.1	49,143
LSTM	89.7 ± 0.9	23,939
ResNet	87.6 ± 1.8	324,931
Random Forest	79.7 ± 0.4	–

† MHCflurry evaluated on its own labels scores 100% — an expected ceiling, not evidence of superior performance; the 8.1-point gap to the CNN reflects the information loss of approximating the labeler's decision boundary from finite samples.

Table 2. Deep FFN performance across label sources (accuracy, mean ± SD across five seeds where noted).

Label source	Accuracy (%)	Notes
DFFN on MHCflurry labels	90.6 ± 1.2	5,190 peptides, 5 seeds
PSSM (biophysics)	82.6	separate 15,000-peptide pool, single seed
Random (i.i.d.)	34.5 ± 1.7	chance-level null, 5 seeds

Table 3. Classification accuracy by encoding strategy.

Encoding	Dim	Accuracy (%)	Notes
BLOSUM62	180	92.2 ± 0.4 (CV)	Stable CV generalisation
ESM-2 t6 mean pool	320	65.9	Positional info destroyed
ESM-2 t12 per-pos	4,320	90.9	Overfitting
ESM-2 t12 + L2	4,320	91.5	Partially mitigated
ESM-2 t6 per-pos	2,880	93.3 (82.2 ± 1.0 CV)	Single-split optimistic

Table 4. Epitope-altering cancer hotspot mutations.

Mutation	WT peptide	Mutant peptide	Effect	Δ score
p53 R248W	MNRRPILTI	MNWRPILTI	CREATED	+0.41
KRAS G12V	YKLVVVGAG	YKLVVVGAV	CREATED	+0.48
KRAS G12V	LVVVGAGGV	LVVVGAVGV	ENHANCED	+0.31
KRAS G12C	LVVVGAGGV	LVVVGACGV	ENHANCED	+0.31
KRAS G13D	GVGKSALTI	DVDKSALTI	DESTROYED	−0.47
KRAS A146T	GIPFIETSA	GIPFIETST	DESTROYED	−0.42
p53 R249S	GMNRRPILT	GMNRSPILT	DESTROYED	−0.16

Table 5. IEDB benchmark performance.

Metric	Value
Sensitivity	95.8% (46/48)
NetMHCpan-4.1 sensitivity	83.3% (40/48)
Specificity	75.0% (15/20)
Precision	90.2%
F1	0.929
ROC AUC	0.953

Table 6. Rebalanced pan16 non-canonical sensitivity by allele (15-allele panel, @0.95).

Allele	Sens	Allele	Sens
A*02:01	0.88	A*11:01	1.00
A*01:01	0.80	A*24:02	0.76
A*03:01	1.00	B*08:01	0.88
B*07:02	0.64	B*15:01	0.96
B*44:03	1.00	B*27:05	0.08
		B*35:01	0.84
		B*40:01	1.00
		B*49:01	1.00
		B*51:01	0.52
		B*57:01	0.92
Pooled	**0.819**		

Table 7. Non-canonical (Type B) sensitivity by predictor class.

Predictor	Canonical	Type B
Deep FFN (sequence)	0.979	0.607
MHCflurry (sequence)	0.986	0.668
MUNIS (sequence)	0.950	0.647

three predictors, despite experimental evidence of peptide-induced HLA-A*02:01-restricted cytotoxic T-cell responses — albeit the weakest of the three mutants screened in that study (Wu *et al.*, 2020). Note that Table 4's "CREATED" label for KRAS G12V refers to the motif-level creation of a canonical P9 anchor (G→V) relative to wild-type, not to a predicted binder: the mutant's absolute score remains below the binding threshold owing to the non-canonical P2-Lys.

3.8 Learned feature representations

Feature extraction from the Deep FFN penultimate dense layer revealed biologically meaningful structure (Figure 6). t-SNE visualisation of its 45-dimensional penultimate representation showed partial class separation: SB peptides clustered distinctly (91% of SB peptides fell within the SB cluster), while WB peptides formed a continuous transition zone between SB and NB clusters — consistent with the graded affinity boundary in the 50–500 nM range. Integrated-gradients analysis identified position-specific anchor contributions: p9 dominated SB predictions, p2 and p3 contributed most to WB predictions, and NB predictions were driven by the absence of anchor-characteristic patterns. Hierarchical clustering separated validated epitopes into coherent clusters, with false negatives forming a distinct branch characterised by absent anchor-position activation. Five features showed Pearson $r > 0.7$ with binding score; by a 1,000-iteration permutation test, approximately 0.3 features would exceed this threshold by chance, suggesting these represent genuine binding-relevant representations rather than spurious correlations.

4 Discussion

4.1 Methodological warnings

Three apparent findings in this project dissolved under controlled re-analysis, each illustrating how benchmark composition can manufacture false conclusions. First, an apparent per-allele superiority over a pan model was a 10.6× training-data-size artifact, not biology. Second, a claimed "label quality dominates architecture" result collapsed once the degenerate random-label null (which performed at chance, ∼33%) was

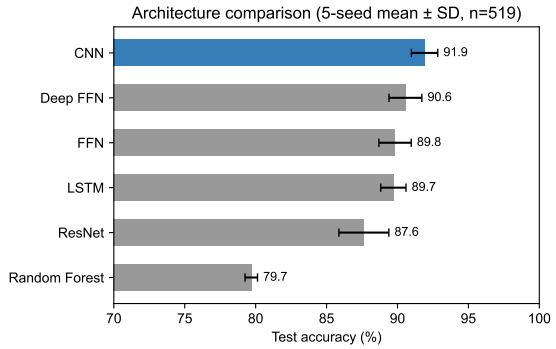


Fig. 1. Model performance comparison across six architectures on the MHCflurry-labelled test set (accuracy, macro F1, WB F1).

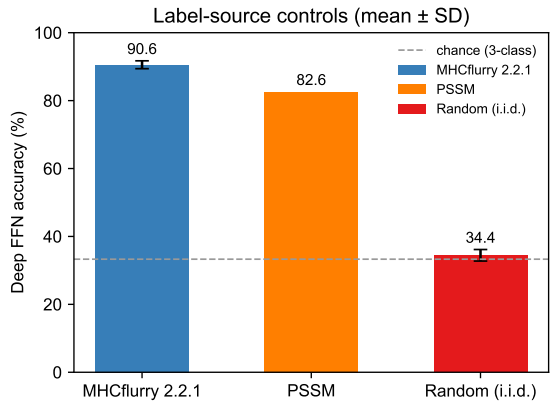


Fig. 2. Architecture comparison (Table 1) and label-source controls: MHCflurry (90.6%), PSSM (82.6%), random labels at chance (∼33%).

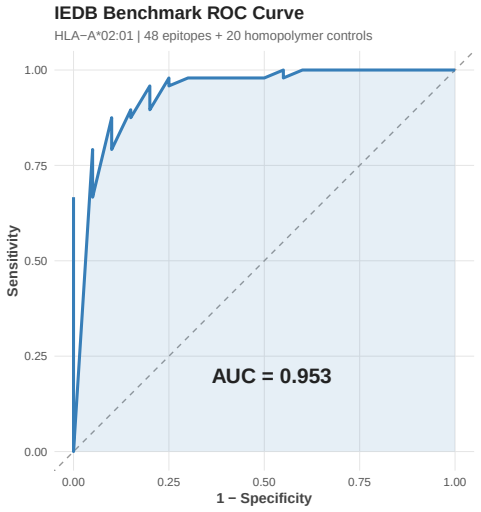


Fig. 3. IEDB benchmark: ROC curve (AUC 0.953), confusion matrix, per-epitope scores.

removed — the realistic label effect is smaller than the architecture range. Third, a reported "simple architecture beats complex ones" ranking was an artifact of stale model checkpoints: re-training with fixed seeds showed a CNN instead modestly outperforms the Deep FFN. We recommend that

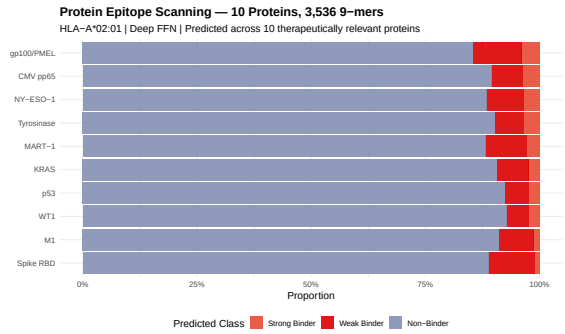


Fig. 4. Protein epitope scanning: SB/WB/NB distribution across 10 proteins and top epitope candidates.

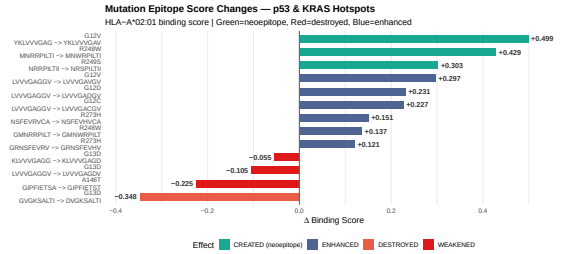


Fig. 5. Cancer hotspot mutation analysis: delta scores and wild-type/mutant pairs for 7 epitope-altering mutations.

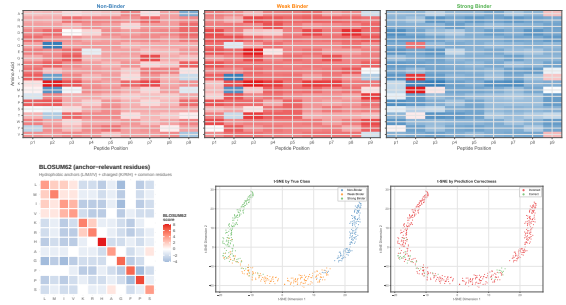


Fig. 6. Learned feature representations: integrated-gradients anchor attention by class, anchor-relevant BLOSUM62 residues, t-SNE by class, and t-SNE by prediction correctness.

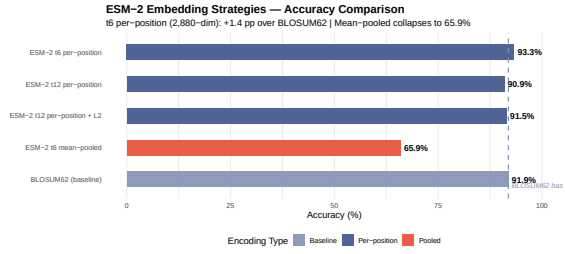


Fig. 7. ESM-2 embedding comparison across five encoding strategies.

peptide-MHC benchmarks report multi-seed estimates with confidence intervals, protein-background negatives, frozen data snapshots, and a state-of-the-art comparator such as NetMHCpan, to avoid these failure modes. The same failure modes recur beyond peptide-MHC: data leakage from protein language-model pretraining distorts downstream

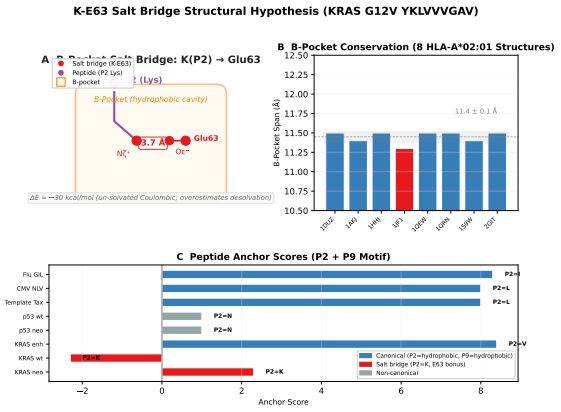


Fig. 8. K-E63 salt-bridge structural hypothesis for KRAS G12V (YKLVVVGAV) presentation.

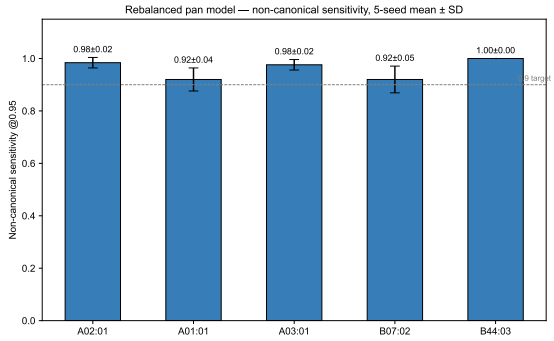


Fig. 9. Multi-allele generalization: rebalanced pan model non-canonical sensitivity (5-seed mean ± SD) across five common HLA-I alleles.

task performance whenever train/test splits are not pretraining-aware (Hermann *et al.*, 2024), and leakage is now recognised as a field-wide reproducibility problem affecting hundreds of papers across seventeen fields (Kapoor and Narayanan, 2023). Dedicated guidance now exists for avoiding leakage in biological machine learning (Bernett *et al.*, 2024) and for reporting clinical prediction models (Collins *et al.*, 2024); Box 1 instantiates these general principles for peptide-MHC benchmarking specifically.

Box 1. Benchmark hygiene recommendations. The three artifacts translate into reusable practices for immunoinformatics and machine-learning benchmarks: (1) equalise training-data size across model arms and report per-allele sample counts; (2) use a true i.i.d. random-label null (chance ≈ 33% for three balanced classes) rather than a degenerate control; (3) freeze data snapshots, model checkpoints, and label versions, and report multi-seed mean ± SD with confidence intervals; (4) report label provenance and break label circularity with a temporal holdout and an independent state-of-the-art comparator (NetMHCpan). Each failure mode reproduced a false conclusion that was caught only by controlled re-analysis.

Architecture choice has a modest but measurable effect on peptide-MHC binding prediction. Across five seeds, a CNN (91.9%) outperformed a Deep FFN (90.6%) by ∼1.3 percentage points (not a formal significance test; the gap is within seed-to-seed variability), while a Random Forest was clearly worse (79.7%), for a maximum inter-architecture range of ∼12 percentage points. The top four architectures fell within ∼2 percentage points, indicating that data quality and the decision boundary matter more than architectural complexity for this task. Random-label controls performed at chance (∼33%), confirming that the observed performance reflects genuine signal rather than noise overfitting.

A PSSM-derived label source yielded lower accuracy than MHCflurry labels (82.6% vs 90.6%, though on a separate peptide pool and therefore not a clean comparison). This does not support the hypothesis that deterministic biophysical labels are inherently superior to ML-derived labels; rather, it suggests that MHCflurry's labels are at least as learnable as a simple PSSM. We emphasise that training on any computational label — including MHCflurry — re-deposits predictor bias into downstream models, and recommend that benchmarks report label provenance and the WB-class F1 as a standard diagnostic.

An independent state-of-the-art predictor, NetMHCpan-4.1, agreed with the MHCflurry labels on only 62.24% (323/519) of the same held-out test peptides — substantially below the 90.6% achieved by a model trained directly on those labels. This ∼28-point gap quantifies the label-circularity limitation: our trained models learn MHCflurry's specific decision boundary, whereas an independent predictor disagrees on roughly one peptide in three, showing that the reported accuracies measure fidelity to a particular labeler rather than true binding.

BLOSUM62 proved more robust than ESM-2 at single-allele training scales; high-dimensional per-position embeddings overfit small datasets. The 2,880-dimensional per-position ESM-2 features create an unfavourable feature-to-sample ratio for the 4,671-sample training set, and the single-split 93.3% result reflects a favourable partition rather than a genuine encoding advantage. The shallower 6-layer ESM-2 variant outperformed the deeper 12-layer model, possibly because t12's additional transformer layers encode long-range protein-level interactions that introduce noise at the short 9-mer peptide scale. Architectural complexity beyond depth with batch normalisation gave diminishing returns; a simple PSSM label source yielded lower accuracy (82.6%) than MHCflurry labels on its own pool, showing that deep learning adds value over a hand-crafted scoring matrix for this task. The language-model route may still be viable at single-allele scale if the model is continued-pretrained on HLA-associated peptides rather than used as a frozen embedder, as such pre-training yields the largest affinity gains precisely at small-data alleles (Mares *et al.*, 2025).

The multi-allele extension shows the recipe is portable: one rebalanced pan model matches per-allele specialists on four of five alleles (B*07:02 excepted) and reaches 0.819 pooled sensitivity on 15 alleles, approximating NetMHCpan's practical value with a simple DFFN. This aligns with the pan-specific paradigm: MixMHCpred v3.0's pan-allele model now outperforms state-of-the-art MHC-I ligand predictors across alleles and species (Tadros *et al.*, 2025). Yet non-canonical (motif-failing) binders plateau at ∼0.65 sensitivity across three independent sequence-based predictors: our Deep FFN (0.607), MHCflurry (0.668), and the ESM-2-based MUNIS (0.647) fall within a narrow band (0.61–0.67), far below the ∼0.98 canonical sensitivity, indicating this ceiling is shared across sequence architectures of varying complexity rather than specific to our simple network. The energy-compensated tail (P2-Lys/Arg/His reaching B-pocket Glu63) is prediction-resistant to all three. The three-tier screening pipeline we considered (sequence filtering → structural scoring → energy-minimisation validation) remains a hypothesis for these configurations, pending experimental validation, since the structural analysis presented here (Section 3) is a single-case docking hypothesis rather than a scaled benchmark. AlphaFold3 now models pMHC geometry well (mean DockQ ≈0.54 on TCR–pMHC complexes), yet its confidence metrics are poor predictors of binding affinity, so structure should complement rather than replace sequence scoring for the hardest cases (Ascunce-París *et al.*, 2026). The canonical binder problem is effectively solved; the non-canonical tail defines the current frontier. This frontier extends beyond anchor-motif violations to non-coding, fusion-derived,

and post-translationally modified peptides, for which current predictors remain under-powered (Vo *et al.*, 2025).

4.2 Translational implications

The cancer neoepitope candidates identified here are testable translational hypotheses. p53 R248W, which created MNWRPILTI, is among the most frequent TP53 missense mutations across human cancers (∼3.5% of somatic TP53 mutations, IARC) with documented gain-of-function oncogenic activity. A vaccine targeting this neoepitope could, only if immunogenicity is experimentally confirmed, apply across multiple tumour types; validation (e.g., IFN-γ ELISpot using HLA-A*02:01+ donor PBMCs) is required before any clinical development. KRAS G12C, which enhanced an existing epitope (WB→SB), is the druggable mutation targeted by sotorasib and adagrasib, suggesting a rational combination immunotherapy strategy. The three epitope-destroying mutations (KRAS G13D, KRAS A146T, p53 R249S) represent potential immune-evasion mechanisms. We note that binding prediction alone does not establish immunogenicity: prior work shows that only a small fraction of top-predicted MHC-I binders are confirmed immunogenic, and that peptide–MHC stability predicts immunogenicity better than affinity (Harndahl *et al.*, 2012; Backert and Kohlbacher, 2015). High-throughput thermostability profiling now quantifies pMHC stability at immunopeptidome scale and improves neoepitope immunogenicity prediction (Jappe *et al.*, 2020), and transfer learning further improves kinetic-stability prediction from limited data (Fasoulis *et al.*, 2024). The binding-to-immunogenicity gap remains the central bottleneck in translating computational epitope prediction to clinical vaccine design.

4.3 Limitations

Training labels are MHCflurry-derived rather than experimental affinities; we quantified the circularity by cross-referencing the 49 curated IEDB epitopes against MHCflurry's training corpus — 44/49 (89.8%) overlapped, a figure computed on all 49 before the exclusion of ILRGSVAHK (HLA-B*08:01) that reduces the evaluated set to 48; all five non-overlapping epitopes were nonetheless correctly classified (5/5, 100%), providing limited evidence of generalisation beyond the training distribution. The single-allele design was deliberate — holding the allele constant isolates label-source variation from allele-driven confounds — but the multi-allele extension addresses the generalisability question directly. The multi-allele estimates rest on IEDB-derived positives; experimental (non-IEDB) validation of the extreme tail is outstanding, and a pMHC-fine-tuned structure model (beyond generic AlphaFold-Multimer) might lift the Type B plateau. The K-E63 salt bridge, all translational predictions, and the cancer neoepitope candidates require experimental confirmation (e.g., IFN-γ ELISpot or tetramer staining). Antigen-processing steps (proteasomal cleavage, TAP transport) are not modelled.

5 Conclusion

Architecture choice has a modest effect on peptide-MHC binding prediction: a CNN modestly outperformed a Deep FFN, the top four architectures fell within ∼2 percentage points, and a Random Forest was clearly worse. Random-label controls confirmed chance-level performance, ruling out noise overfitting. Canonical binding is largely solved by the anchor motif — a zero-parameter anchor rule matches the trained model — while non-canonical binders plateau at ∼0.65 sensitivity across three independent sequence-based predictors, defining the current frontier. Beyond peptide-MHC, the three artifacts we corrected — data-size imbalance, a degenerate null, and stale checkpoints — illustrate a general benchmark-composition failure mode that reproduces false conclusions across machine-learning benchmarks; the checklist in Box 1 is offered as a reusable safeguard. Experimental validation of the non-canonical tail is the decisive next step.

Author contributions

Yunshi Huang and Weisheng Chen: conception, design, data analysis, manuscript writing. Yiqi Cai: data analysis, manuscript writing. Xiaopeng Zhuang: data interpretation, manuscript revision. Zhuxian Zhou: data interpretation, manuscript revision. Xiangyang Xue and Zhuha Zhou: supervision, manuscript revision. All authors read and approved the final manuscript.

Competing interests

The authors declare no competing interests.

Data and code availability

All code, trained models (per-allele `deep_ffn_HLA-A_*.keras` and pan `deep_ffn_pan_seed*.keras`/`deep_ffn_pan16.keras`), feature matrices, prediction results, the curated multi-allele dataset (`type_ab_dataset.json`, 14,488 positives), and the experimental-label benchmark (6,524 measured IC50 values) are available at github.com/woodhaha/peptide-mhc-benchmark. The per-allele Deep FFN is distributed as the `pmhcbenchmark` Python package, with a `pmhc-predict` command-line tool and a standalone Windows GUI (`pmhc_predict.exe`); exported weights (`pmhc_models.npz`) enable tensorflow-free inference. The single-allele benchmark-hygiene analysis is the primary contribution of this manuscript; the multi-allele benchmark reported in Sections 3.6–3.7 is shared with a companion manuscript submitted separately to *PLOS Computational Biology*, which presents the pan-vs-per-allele comparison in full, with no duplicated figures or tables between the two papers.

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Ethics statement

This study used only publicly available, de-identified data from the Immune Epitope Database (IEDB) and public protein databases; no human subjects, animals, or patient-derived samples were involved, and no ethics approval or informed consent was required.

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